

Estimating Pi with Random Points: A Monte Carlo Experiment

SimuPhysics Interactive Pi Estimation (Monte Carlo) · SimuPhysics

Year 10	~30 minutes	Mathematics · Statistics & Probability
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NOTICE — AI-generated worksheet

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Simulation: <https://simuphysics.com/view/mathematics-interactive-pi-estimation-monte-carlo.html>
Curriculum links: AC9M10P02 · AC9M10ST01

Learning intentions

- I can describe how a Monte Carlo simulation uses randomness to estimate a value.
- I can design and run repeated trials and record how results change with sample size.
- I can evaluate how sample size affects the accuracy and reliability of a simulated estimate.

Getting started

Open the simulation and start with a small number of random points (e.g. 100). Note the estimate of π it produces, then work through the tasks below, increasing the number of points each time you're asked to.

Tasks

1. With 100 points, record the simulation's estimate of π and how far off it is from the true value (3.14159...).

2. Run the simulation at three increasing point counts (e.g. 100, 1000, 10000), recording the π estimate each time.

Number of points	Estimated value of π

3. Describe the trend you see in your table as the number of points increases.

4. Using the ratio of points landing inside the circle to the total number of points, explain why more random points produce a more accurate estimate of π .

5. Suppose the square has side length 2 units. Calculate the area of the square and the area of the inscribed circle, then explain why $(\text{points inside circle} \div \text{total points}) \times \text{square area}$ should approximate the circle's area — and how this leads to an estimate of π .

Working:

6. Based on your table, evaluate whether a single run of this simulation with a small number of points is a reliable way to estimate π .

7. Describe one real-world situation where the same idea — more trials producing a more reliable estimate — also applies.

Extension (optional)

Mathematicians have other ways to calculate π that don't rely on randomness (for example, geometric or series-based methods). Briefly compare how reliable you'd expect one of these methods to be, at a fixed amount of computing effort, compared with the Monte Carlo method here.
